AI IN HEALTHCARE – INTERNAL BRIEF FOR PILOT DEMO

1. Executive Summary

   Healthcare is under massive pressure: rising costs, staff burnout, long diagnosis cycles, and inconsistent patient outcomes. AI is finally moving from hype to production. The first real, defensible use cases are not surgery robots — they are decision support, workflow automation, and early risk prediction.

This brief summarizes:

- Where AI is actually working today
- What business value it delivers
- What risks leadership cares about

Key points:

- AI is not replacing doctors. It's reducing repetitive manual work and catching issues earlier.
- Highest ROI areas right now:

  • Radiology triage

  • Hospital operations (bed management, staffing)

  • Revenue cycle automation (coding, prior auth, denial management)

- GenAI "copilots" are starting to cut clinical documentation time by 30–40%.
- The main blockers are not the models — it's data quality, privacy, and workflow adoption.

2. Core AI Use Cases (2025 Reality)

   2.1 Diagnostic Support (Radiology, Pathology, Cardiology)

- AI models flag high-risk scans (stroke, internal bleeding, etc.) so radiologists can prioritize.

- This is triage support, not final diagnosis.
- Business impact: Faster ER turnaround, fewer missed critical cases.
- Risk: Regulatory approval, liability, clinician trust.

2.2 Clinical Documentation Copilots

- Ambient listening + GenAI turns doctor–patient conversation into structured notes in the EHR.
- Value: Doctors spend less time typing and more time with the patient.
- Metric: Time-per-encounter (some pilots show 28%–43% reduction).

2.3 Predictive Risk Scoring (Readmission, Sepsis, Falls)

- ML models analyze vitals, labs, and medication history to assign risk scores.
- Example: “High sepsis likelihood in next 6 hours” alert to ICU team.
- Value: Preventable ICU escalations = cost savings + better patient outcomes.
- Risk: Alert fatigue — if everything is critical, nothing is critical.

2.4 Operational / Financial Automation

- Coding, prior auth, denial management, insurance language translation.
- Value: Direct revenue protection. Often the first AI use case a hospital approves because ROI is easy to show.

3. Data Stack Reality

   Typical hospital data sources:

- EHR systems (Epic, Cerner): notes, orders, vitals, meds
- PACS: imaging archives (CT, MRI, X-ray)
- Billing / claims / rev cycle systems

Main problem: data is siloed, inconsistent, and stored in 20+ formats.

Things leadership must hear:

- PHI/HIPAA: You cannot leak unredacted data into a public LLM.
- Auditability: Legal + clinicians need to understand WHY the model made a call.
- Drift: A model trained on last year's population might fail in a new demographic.

If you try to "do AI" without solving data integration + governance first, you get noisy results and legal risk.

4. ROI Model (Clinical Documentation Copilot Example)

   Baseline:

- Avg provider spends ~2.2 hours/day on charting.
- Fully loaded clinician cost: about $120/hour.

Impact:

- AI copilot cuts charting time by ~40% (real pilots: 28–43%).
- That's ~0.9 hours saved per clinician per day ≈ $108/day.
- Across 120 clinicians, that's ~$12,960/day → about $3.7M/year.

Hidden value:

- Lower burnout → less turnover → less expensive contract staffing.
- Cleaner structured notes → better billing codes → fewer denied claims.

5. Risk & Compliance Notes for Leadership

   To get executive buy-in, you need to say clearly:

- Is AI giving a suggestion or making a decision?
- Are we logging who saw an alert and who acted?
- Can we turn it off fast if it drifts?
- How are we making sure it's not biased (ex: missing stroke risk in women or people of color)?
- Can we explain the reasoning to auditors/regulators?

6. How to Design a Safe First Pilot

Phase 1: Narrow problem, measurable outcome

- Example: “Reduce ER CT-scan read time by 20%.”
- Do NOT start with “AI transformation of the entire hospital.”

Phase 2: Human-in-the-loop

- AI flags high-priority scans.
- Radiologist still makes the call.
- Everything is tracked.

Phase 3: Executive dashboard

Show weekly:

- Avg turnaround time
- % false positives
- Clinician satisfaction
- And most importantly: “This saved X hours / Y dollars / Z escalations.”

This is what convinces leadership.

7. Questions Executives Always Ask

Q1. Will this replace doctors?

A1. No. Early phase AI is “assistive,” not autonomous. Final decision stays with the clinician.

Q2. What if it’s wrong and someone dies?

A2. That’s why all clinical use is human-in-the-loop and legally approved escalation paths.

Q3. Do we need to train our own model from scratch?

A3. Usually no. You fine-tune or do retrieval (RAG) on internal data. Full pretraining is not realistic for hospitals.

Q4. How fast can we show value?

A4. Revenue cycle automation: weeks.

Clinical safety support: longer, because it touches patient care.